

Shubhadeep Mandal

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EDUCATION

International Institute of Information Technology, Hyderabad <i>Telangana, India</i> <i>Master of Technology in Computer Science and Engineering</i>	July 2025 – Present <i>CGPA: 9.12</i>
Indian Institute of Information Technology, Ranchi <i>Ranchi, India</i> <i>Bachelor of Technology (Hons.) in Computer Science and Engineering</i>	Aug 2020 – Jun 2024 <i>CGPA: 9.05</i>

PROJECTS

Harpy: AI Competitive Programming Toolkit <i>Python, MCP Server, C++, Subprocesses, CI/CD</i> 🎧
– Architected an open-standard JSON-RPC 2.0 stdio MCP server exposing 4 deterministic tools, connecting LLM coding agents (Cursor, Antigravity, Claude) to local C++ compiler toolchains and CPH test runners. – Engineered automated brute-force differential testing and a sandboxed multi-language runner with sub-millisecond execution benchmarking, TLE watchdog interrupts, and side-by-side terminal diffs. – Implemented dual-channel CPH synchronization via raw HTTP sockets and offline .cph test files, distributed as a PyPI package and PyInstaller standalone binary backed by Python 3.10–3.12 CI.
IndicLLM: Transformer Pretraining & Reasoning SFT <i>PyTorch, FlashAttention-2, RoPE</i> 🎧
– Pretrained 25M-param decoder-only Transformers from scratch on 1B+ tokens across Hindi and Assamese using bare PyTorch primitives, driving a +36.5% perplexity reduction via FlashAttention-2, RoPE, and SwiGLU. – Trained dedicated 16k SentencePiece BPE tokenizers on AI4Bharat Sangraha, eliminating token fertility inflation and achieving 0.505 Bits-Per-Byte sequence compression across both high and low-resource languages. – Fine-tuned 8 model variants across 20k synthetic multi-hop reasoning tasks, unlocking a +45.8% relative Token F1 gain and 61.3% decomposed CoT step score over direct SFT via intermediate deductive scratchpads.
BoardBuddy: Real-Time Collaborative Canvas <i>Next.js 14, Convex DB, Liveblocks, TypeScript</i> 🎧
– Architected a collaborative workspace supporting real-time multi-user syncing, freehand sketching, and vector shapes. – Synchronized real-time canvas state and user presence across concurrent client sessions using Liveblocks and Convex DB. – Secured organization-scoped workspaces and role permissions with Clerk authentication and a responsive TailwindCSS UI.
Peer-to-Peer Distributed File Sharing System <i>C++, POSIX Sockets, Multithreading, OpenSSL, Linux</i> 🎧
– Engineered a BitTorrent-style hybrid P2P system in C++, achieving an 8.5x transfer speedup on 1GB files (193s to 22.8s) across 5 concurrent peers by saturating network and disk I/O. – Implemented out-of-order parallel chunk writing directly to designated file offsets and dynamic bitfield generation for real-time partial seeding during active downloads. – Architected Primary-Backup tracker failover with 2s bounded failover, full state synchronization, and verified resilience via 48 unit tests under AddressSanitizer/UBSan.

TECHNICAL SKILLS

Languages: C/C++, Rust, Python, TypeScript, JavaScript, SQL, Bash.
AI Systems & LLMs: Model Context Protocol (MCP), Transformers (Pretrain & SFT), FlashAttention-2, RoPE, PyTorch.
Systems & Tooling: Subprocess Sandboxing, POSIX Sockets, Multithreading, Linux/Unix, Docker, Git, CI/CD.
Full-Stack & Cloud: Next.js 14, React, Node.js, Convex DB, Prisma ORM, PostgreSQL, WebSockets, REST APIs.

ACHIEVEMENTS

- **B.Tech Rank Holder:** Ranked 2nd in IIIT Ranchi by GPA in B.Tech CSE.
- **PGEE 2025:** Secured All-India Rank 100 in IIIT-Hyderabad Post Graduate Entrance Examination (PGEE).
- **Megathon 2025:** Third rank team in Megathon 2025 in the mobile computing track.

INTERESTS

- Web Novel and Manga Enthusiast.
- Chess and Badminton.